

Diagnostic Goodness-of-Fit Tests for Incidence and Latency
Components in Mixture Cure Models with Application to
Pediatric Acute Myeloid Leukemia

A Thesis

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Abstract

Mixture cure models are widely used in survival analysis to account for a cured fraction among patients. These models separate the patients into two components: an incidence component indicating the probability of being susceptible (not cured), and a latency component describing the survival distribution of susceptible patients. Although mixture cure models have been successfully applied in many studies, the validity of the assumed parametric forms for either component has received limited attention. Therefore, we implemented two goodness-of-fit tests for mixture cure models. The first test, proposed by Muller and Van Keilegom (2019), evaluates the correctness of the parametric form of the incidence component. The second test, proposed by Geerdens, Janssen and Van Keilegom (2020), evaluates the correctness of the parametric form of the latency component. We applied both diagnostic tests to the TARGET-AML dataset, focusing on pediatric acute myeloid leukemia patients with an MLL rearrangement. We then fitted a mixture cure model using clinical covariates from the MLL dataset to evaluate association with the incidence and latency components of the mixture cure model.

This is dedicated to my advisor, for her consistent support, guidance, and kindness.

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Table of Contents

	Page
Abstract	ii
Dedication	iii
Acknowledgments	iv
Vita	v
List of Tables	ix
List of Figures	x
1. Introduction	1
1.1 Organization of this Thesis	3
2. Mixture Cure Model	4
2.1 Model Framework	4
2.2 Likelihood Under Right Censoring	4
2.3 Incidence Component	5
2.4 Latency Component	6
3. Goodness-of-Fit Test for the Incidence Component	7
3.1 Introduction	7
3.2 Hypotheses	8
3.3 Nonparametric Estimation of the Cure Probability	9
3.4 Fitting the Parametric Incidence Model	9
3.5 Test Statistic	10
3.6 Bootstrap Procedure for the P-value	10

3.7	Bandwidth Choice	12
3.8	Simulation Results	12
4.	Goodness-of-Fit Test for the Latency Component	16
4.1	Introduction	16
4.2	Hypotheses	16
4.3	Estimating the Susceptible Fraction	17
4.4	Nonparametric Estimation of the Susceptible Survival Function . .	17
4.5	Fitting the Parametric Latency Model	18
4.6	Test Statistic	18
4.7	Bootstrap Procedure for the P -value	19
4.8	Simulation Results	20
5.	Application to the Pediatric AML MLL Subgroup	26
5.1	Introduction	26
5.2	Data Description	26
5.3	Mixture Cure Model Fitting	27
5.4	Application Results	28
6.	Discussion	32
	Appendices	34
A.	R Code for Incidence Goodness-of-Fit Test	34
A.1	Simulation Function	34
A.2	Nonparametric Estimation of Cure Probability	35
A.3	Logistic Estimation	35
A.4	Test Statistic	36
A.5	Incidence Goodness-of-Fit Test	36
A.6	Simulation Study for the Incidence Goodness-of-Fit Test	39
B.	R Code for Latency Goodness-of-Fit Test	42
B.1	Simulation Function	42
B.2	Estimate of the Susceptible Fraction	43
B.3	Nonparametric Latency Survival Estimate	43
B.4	Parametric Latency Model Fitting	44
B.5	Latency Goodness-of-Fit Test Statistic	49

B.6	Bootstrap Latency Goodness-of-Fit Test	49
B.7	Simulation Study for the Latency Goodness-of-Fit Test	52
C.	R Code for the Application to the MLL Dataset	55
C.1	Load Packages and Data	55
C.2	Kaplan–Meier Plot	55
C.3	Gene Expression Data Cleaning	56
C.4	Phenotypic Data Cleaning	56
C.5	Final Analysis Dataset	57
C.6	Data Summary	58
C.7	Model 1: Weibull Mixture Cure Model	59
C.7.1	Model Fitting	59
C.7.2	Incidence Goodness-of-Fit Test	59
C.7.3	Latency Goodness-of-Fit Test	60
C.8	Model 2: Semiparametric Mixture Cure Model	60
C.8.1	Model Fitting	60
C.8.2	Incidence Goodness-of-Fit Test	60
	Bibliography	62

List of Tables

Table		Page
3.1	Rejection proportion (%) of the incidence GOF test statistic $\tilde{T}_n$ for different values of c_h , c_T , θ_2 , and sample size n . The simulation setting is $\theta_0 = 1$ and $\theta_1 = 1$. The null hypothesis corresponds to $\theta_2 = 0$, and the significance level is $\alpha = 0.05$	15
4.1	Simulation results for the latency goodness-of-fit test under a long plateau scenario ($\phi = 0.75$). Each result is based on 1000 simulated datasets with 500 bootstrap samples. Rejection rates (%) at the 5% and 10% significance levels are reported.	23
4.2	Simulation results for the latency goodness-of-fit test under a short plateau scenario ($\phi = 0.25$). Each result is based on 1000 simulated datasets with 500 bootstrap samples. Rejection rates (%) at the 5% and 10% significance levels are reported.	24
4.3	Comparison of rejection rates between the results reported in Geerdens et al. and the results obtained in this thesis. Rejection rates (%) at the 5% and 10% significance levels are shown under the long plateau ($\phi = 0.75$) and short plateau ($\phi = 0.25$) settings.	25
5.1	Incidence goodness-of-fit test for the logistic incidence model under two mixture cure models applied to the Pediatric AML MLL data. Both models use a logistic specification for the incidence component.	30
5.2	Results of the latency goodness-of-fit test applied to the Pediatric AML MLL data under different parametric latency distributions. The table reports the test statistic, bootstrap p -value, and the Akaike Information Criterion (AIC) for each fitted latency model.	31

List of Figures

Figure		Page
5.1	Kaplan–Meier survival curve for the Pediatric AML MLL subgroup. The yellow shaded region represents the 95% confidence interval, and tick marks indicate censored observations. The curve shows a plateau after approximately 5 years, suggesting the presence of a cure fraction.	29
5.2	Kaplan–Meier survival curve for the Pediatric AML MLL subgroup overlaid with fitted survival curves under Weibull, Gompertz, lognormal, and uniform mixture cure models. Tick marks indicate censored observations.	30

Chapter 1: Introduction

In survival analysis, it is common to assume that all patients will eventually experience the event if they are followed long enough. However, in some time to event analyses, a portion of patients may never experience the event of interest. For example, when studying recurrence, some patients may be cured and therefore never experience the event, and when studying disease-free survival, some patients may never relapse. In this case, subjects who will not experience the event of interest can be considered cured. Therefore, standard survival models such as the Cox proportional hazards model may not be appropriate when a cured fraction is present.

Pediatric acute myeloid leukemia (AML) is one example where long-term survivors exist. With improved treatment, some patients remain event-free for many years. As a result, when examining the Kaplan–Meier curve for these patients, the survival curve often shows a plateau after several years of follow-up, and this pattern suggests the presence of a cure fraction.

To address this issue, mixture cure models were developed by Berkson and Gage (1952) [2, 3]. These models separate patients into two groups: cured and susceptible. Specifically, the model contains two parts. The first part is the incidence component, which models the probability of being susceptible. The second part is the latency component, which models the survival distribution among susceptible patients. In

this way, factors associated with cure probability and factors associated with time to event can be studied separately, as first proposed by Farewell (1977) [6].

Although mixture cure models are widely used, the parametric assumptions in the incidence and latency components are usually not carefully checked [1]. Nevertheless, if these assumptions are incorrect, the results may be biased. Therefore, it is important to apply appropriate methods to assess whether the chosen parametric forms are appropriate.

In this thesis, two goodness-of-fit tests for mixture cure models are implemented. First, the test proposed by Müller and Van Keilegom (2019) [9] evaluates the parametric form of the incidence component. Second, the test proposed by Geerdens, Janssen, and Van Keilegom (2020) [7] evaluates the parametric form of the latency component. In addition, simulation studies are conducted to examine the performance of these tests under finite samples. Finally, the methods are applied to a real dataset.

The main objectives of this thesis are as follows:

1. To implement the goodness-of-fit test for the incidence component of a mixture cure model.
2. To implement the goodness-of-fit test for the latency component.
3. To evaluate the performance of these tests using simulation studies.
4. To apply the tests to the pediatric AML MLL dataset and interpret the results.

1.1 Organization of this Thesis

The rest of this thesis is organized as follows: Chapter 2 introduces the mixture cure model, including model formulation and estimation under right censoring; Chapter 3 describes the goodness-of-fit tests for the incidence component. In addition, the simulation study design and results are presented; Chapter 4 describes the goodness-of-fit tests for the latency component. In addition, the simulation study design and results are presented; Chapter 5 presents the application of the methods to the Pediatric AML MLL dataset and reports the diagnostic results; Chapter 6 summarizes the findings, limitations, and discusses future research directions.

Chapter 2: Mixture Cure Model

A mixture cure model is used when there are patients who may be considered cured. The model consists of two parts: the incidence component (cured patients), and the latency component (susceptible patients who are at risk of experiencing the event). Therefore, the overall survival function can be written as a combination of the cure probability and the survival distribution among the susceptible patients, as shown in Equation (2.1).

2.1 Model Framework

Let T be the event time and let $\mathbf{X}$ be a vector of covariates. In a mixture cure model, the survival function of T given $\mathbf{X}$ is written as

$$S(t | \mathbf{X}) = P(T > t | \mathbf{X}) = p(\mathbf{X}) + \{1 - p(\mathbf{X})\}S_1(t | \mathbf{X}) \quad (2.1)$$

Where $p(\mathbf{X})$ is the cure probability given covariates $\mathbf{X}$, $1 - p(\mathbf{X})$ is the probability of being susceptible, and $S_1(t | \mathbf{X})$ is the survival function for the susceptible patients.

2.2 Likelihood Under Right Censoring

In practice, survival data are usually right censored. Let C be the censoring time, and Y be the observed time. We observe

$$Y = \min(T, C), \quad \delta = I(T \leq C),$$

where $\delta = 1$ indicates an event and $\delta = 0$ indicates censoring. For subject i , the observed data are $(Y_i, \delta_i, \mathbf{X}_i)$.

Under the mixture cure model, the density of T among susceptible patients is

$$f_1(t | \mathbf{X}) = -\frac{d}{dt}S_1(t | \mathbf{X}).$$

The contribution to the likelihood for a given subject then depends on whether the event was observed or censored. If $\delta = 1$, then an event is observed, which implies the subject is susceptible. Therefore, the likelihood contribution is

$$L(\delta = 1) = \{1 - p(\mathbf{X})\}f_1(Y | \mathbf{X}). \quad (2.2)$$

If $\delta = 0$, then the subject is censored at time Y . In this case, the subject may be cured, or may be susceptible but has not had the event by time Y . Therefore, the likelihood contribution is

$$L(\delta = 0) = p(\mathbf{X}) + \{1 - p(\mathbf{X})\}S_1(Y | \mathbf{X}). \quad (2.3)$$

Combining both cases, the full likelihood contribution can be written as

$$L = [\{1 - p(\mathbf{X})\}f_1(Y | \mathbf{X})]^\delta [p(\mathbf{X}) + \{1 - p(\mathbf{X})\}S_1(Y | \mathbf{X})]^{1-\delta}. \quad (2.4)$$

2.3 Incidence Component

The incidence component models the cure probability $p(\mathbf{X})$, which is commonly modeled using regression

$$\text{logit}(p(\mathbf{X})) = \beta_0 + \mathbf{X}^\top \boldsymbol{\beta} \quad (2.5)$$

where β_0 is an intercept and $\boldsymbol{\beta}$ is a vector of regression coefficients. Under this model, a positive coefficient indicates a higher cure probability or equivalently a lower susceptibility probability.

2.4 Latency Component

The latency component models the survival distribution among susceptible patients. In this thesis, the main model assumes a Weibull distribution for the susceptible event time. Let $S_1(t)$ denote the susceptible survival function. Following Geerdens et al., the Weibull model is written as

$$S_1(t) = \exp(-(\lambda t)^\rho), \quad (2.6)$$

where $\lambda > 0$ and $\rho > 0$. The corresponding density is

$$f_1(t) = \lambda \rho (\lambda t)^{\rho-1} \exp(-(\lambda t)^\rho) \quad (2.7)$$

and the hazard function is

$$h_1(t) = \frac{f_1(t)}{S_1(t)} = \lambda \rho (\lambda t)^{\rho-1}. \quad (2.8)$$

Since the parameterization is different from the default Weibull parameterization used in **R**. In **R**, the Weibull survival function is commonly written as

$$S_1(t) = \exp \left\{ - \left(\frac{t}{b} \right)^\rho \right\},$$

where $b > 0$ is the scale parameter. The two forms are equivalent, with the relationship

$$b = \frac{1}{\lambda}.$$

Therefore, when simulating Weibull event times in **R** using `rweibull()`, the shape parameter is set to ρ and the scale parameter is set to $1/\lambda$ so that the simulated data match the parameterization used in Geerdens et al.

Chapter 3: Goodness-of-Fit Test for the Incidence Component

3.1 Introduction

In a mixture cure model, the incidence component models the cure probability $p(\mathbf{X})$, while $1 - p(\mathbf{X})$ represents the probability of being susceptible. In practice, we usually assume a parametric form for $p(\mathbf{X})$, such as a logistic regression model. However, even when the latency distribution is reasonably flexible, a misspecified incidence model may still lead to incorrect conclusions about the cure fraction. Therefore, it is important to check whether the assumed parametric model for $p(\mathbf{X})$ fits the data well.

In this chapter, we describe the goodness-of-fit test proposed by Müller and Van Keilegom (2019), which tests whether the cure probability as a function of covariates follows a specified parametric model. The main idea of this test is to compare a nonparametric estimator of the cure probability with a fitted parametric model. A bootstrap procedure is then used to approximate the distribution of the test statistic and compute the corresponding p -value [9].

3.2 Hypotheses

Let $\mathbf{X}$ denote a vector of covariates, and let $p(x)$ denote the cure probability at a given covariate value x . The goal is to test whether the cure probability follows a specified parametric model. The null hypothesis is

$$H_0 : \quad p(x) = p_\theta(x), \quad (3.1)$$

and the alternative hypothesis is

$$H_1 : \quad p(x) \neq p_\theta(x). \quad (3.2)$$

Following Müller and Van Keilegom (2019), we consider a binary indicator Z , where $Z = 1$ indicates a susceptible (uncured) subject and $Z = 0$ indicates a cured subject. Under this setup, the cure probability is

$$p(x) = P(Z = 0 \mid \mathbf{X} = x),$$

and the susceptibility probability is

$$1 - p(x) = P(Z = 1 \mid \mathbf{X} = x).$$

Under the null hypothesis, the susceptibility probability is modeled using a logistic regression model,

$$P(Z = 1 \mid \mathbf{X} = x) = \frac{\exp(\theta_0 + \theta_1 x)}{1 + \exp(\theta_0 + \theta_1 x)}, \quad (3.3)$$

and equivalently, the cure probability can be written as

$$p_\theta(x) = P(Z = 0 \mid \mathbf{X} = x) = \frac{1}{1 + \exp(\theta_0 + \theta_1 x)}, \quad x \in R, \quad (3.4)$$

where $\theta = (\theta_0, \theta_1)^\top$ denotes the vector of model parameters.

3.3 Nonparametric Estimation of the Cure Probability

To construct the goodness-of-fit test, we first need a nonparametric estimate of the cure probability, denoted by $\hat{p}(x)$. We used the estimator introduced by Xu and Peng (2014). The idea is that the cure probability corresponds to the height of the plateau of the conditional survival function [10]. Specifically, the estimator is defined as the conditional survival probability evaluated at the largest uncensored event time $Y_{(n)}^1$:

$$\hat{p}(x) = \hat{S}(Y_{(n)}^1 \mid x), \quad Y_{(n)}^1 = \max_{i:\delta_i=1} Y_i. \quad (3.5)$$

Here $\hat{S}(t \mid x)$ is a conditional survival estimator that incorporates covariate information. For each covariate value x , this gives a corresponding estimate $\hat{p}(x)$, so in practice we obtain a set of estimated cure probabilities $\{\hat{p}(x_i)\}_{i=1}^n$ evaluated at the observed covariate values. In practice, $\hat{S}(t \mid x)$ is constructed using a kernel-based extension of the Kaplan–Meier estimator, known as the Beran estimator. The estimator assigns kernel weights around the covariate value x , so that observations with covariate values close to x receive larger weights when estimating the conditional survival function.

3.4 Fitting the Parametric Incidence Model

After obtaining the nonparametric cure probability estimates $\hat{p}(x_i)$ at the observed covariate values $x_1, \dots, x_n$, the next step is to fit the parametric incidence model $p_\theta(x)$. Müller and Van Keilegom (2019) suggest estimating the parameter vector θ by maximizing a Bernoulli-type pseudo log-likelihood based on the pairs $(x_i, \hat{p}(x_i))$.

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^n [\hat{p}(x_i) \log p_{\theta}(x_i) + \{1 - \hat{p}(x_i)\} \log \{1 - p_{\theta}(x_i)\}]. \quad (3.6)$$

3.5 Test Statistic

The incidence goodness-of-fit test statistic measures the difference between the nonparametric estimate $\hat{p}(x)$ and the fitted parametric model $p_{\hat{\theta}}(x)$. Following Müller and Van Keilegom (2019), the test statistic is defined as

$$T_n = nh^{1/2} \int \{\hat{p}(x) - p_{\hat{\theta}}(x)\}^2 \pi(x) dx, \quad (3.7)$$

where $\pi(x)$ is a weight function. In practice, this integral is replaced by an empirical average over the observed covariate values $x_1, \dots, x_n$:

$$\tilde{T}_n = nh^{1/2} \frac{1}{n} \sum_{i=1}^n \{\hat{p}(x_i) - p_{\hat{\theta}}(x_i)\}^2. \quad (3.8)$$

A larger value of $\tilde{T}_n$ indicates a larger discrepancy between the nonparametric estimate and the fitted parametric cure model.

3.6 Bootstrap Procedure for the P-value

Because the asymptotic distribution of the test statistic converges slowly, a test based on the normal approximation may not perform well in finite samples. Müller and Van Keilegom recommend using a model-based bootstrap procedure to approximate the distribution of the test statistic. This approach avoids estimating unknown quantities in the asymptotic variance and provides more accurate inference in moderate sample sizes. The bootstrap procedure is summarized as follows:

1. Using the observed data (Y_i, δ_i, x_i) and a pilot bandwidth h_0 , compute the nonparametric cure estimates $\hat{p}_{h_0}(x_i)$ for $i = 1, \dots, n$.

2. Using $\hat{p}_{h_0}(x_i)$ as pseudo-responses, estimate $\hat{\theta}$ by maximizing the pseudo log-likelihood. Then compute the fitted cure probabilities

$$p_{\hat{\theta}}(x_i) = \frac{1}{1 + \exp(\hat{\theta}_0 + \hat{\theta}_1 x_i)}, \quad i = 1, \dots, n.$$

3. Using the Beran estimator with bandwidth h_0 , estimate the conditional survival function $\hat{S}_{h_0}(t \mid x_i)$ and the conditional censoring survival function $\hat{G}_{h_0}(t \mid x_i)$ for each $i = 1, \dots, n$. Let $\{t_1, \dots, t_m\}$ denote the common time grid returned by the Beran estimator. Then, for each x_i , define the susceptible survival curve by

$$T^*(t \mid x_i) = \frac{\hat{S}_{h_0}(t \mid x_i) - \hat{p}_{h_0}(x_i)}{1 - \hat{p}_{h_0}(x_i)}.$$

4. For $b = 1, \dots, B$, generate a bootstrap sample as follows:

- (a) Generate bootstrap susceptibility indicators $Z_{i,b}^*$ such that

$$P^*(Z_{i,b}^* = 1) = 1 - p_{\hat{\theta}}(x_i),$$

where $Z_{i,b}^* = 1$ indicates a susceptible subject and $Z_{i,b}^* = 0$ indicates a cured subject.

- (b) For each $i = 1, \dots, n$, generate $U_{i,b} \sim \text{Uniform}(0, 1)$ and let $T_{i,b}^*$ be the first time point on the grid $\{t_1, \dots, t_m\}$ such that

$$T^*(t \mid x_i) \leq U_{i,b}.$$

If no such time point exists, set $T_{i,b}^*$ equal to the largest grid value. If $Z_{i,b}^* = 0$, set $T_{i,b}^* = \infty$.

- (c) For each $i = 1, \dots, n$, generate $U_{i,b}^{(c)} \sim \text{Uniform}(0, 1)$ and let $C_{i,b}^*$ be the first time point on the same grid such that

$$\hat{G}_{h_0}(t \mid x_i) \leq U_{i,b}^{(c)}.$$

If no such time point exists, set $C_{i,b}^*$ equal to the largest grid value plus 1.

(d) Construct the bootstrap observed data

$$Y_{i,b}^* = \min(T_{i,b}^*, C_{i,b}^*), \quad \delta_{i,b}^* = I(T_{i,b}^* \leq C_{i,b}^*).$$

(e) Using bandwidth h , recompute the nonparametric cure estimates $\hat{p}_h^*(x_i)$ based on the bootstrap sample and refit the parametric model to obtain $p_{\hat{\theta}_b^*}(x_i)$.

(f) Compute the bootstrap test statistic

$$\tilde{T}_{n,b}^* = nh^{1/2} \left(\frac{1}{n} \sum_{i=1}^n \left\{ \hat{p}_h^*(x_i) - p_{\hat{\theta}_b^*}(x_i) \right\}^2 \right). \quad (3.9)$$

5. Finally, recompute the observed test statistic $\tilde{T}_n$ from the original data using bandwidth h , and calculate the bootstrap p -value as

$$\text{p-value} = \frac{1}{B} \sum_{b=1}^B I(\tilde{T}_{n,b}^* \geq \tilde{T}_n). \quad (3.10)$$

3.7 Bandwidth Choice

Müller and Van Keilegom recommend using two bandwidth parameters: a pilot bandwidth h_0 and a second bandwidth h used in the test statistic:

$$h = c_h n^{-0.2}, \quad h_0 = c_h n^{-0.11},$$

where typical values for c_h include 1, 1.5, and 2.

3.8 Simulation Results

In this simulation study, the covariate X is generated from a uniform distribution on $(-1, 1)$. For each subject, a binary susceptibility indicator Z is then generated from

a logistic model. Under the null hypothesis, the susceptibility probability follows the logistic model in Equation (3.3). We also consider an extended logistic model under the alternative, given by

$$p(x) = \frac{1}{1 + \exp(\theta_0 + \theta_1 x - \theta_2 x^2)}. \quad (3.11)$$

$\theta_2 = 0$ corresponds to the correctly specified logistic incidence model, while $\theta_2 = 1, 2, 4$ introduces increasing levels of misspecification in the incidence model. For susceptible subjects, the latency time is generated from a Weibull distribution with covariate-dependent shape and scale parameters. Specifically, the shape parameter is $a(x) = (1 + x)^{c_T}$ and the scale parameter is $b(x) = \exp\{-(1 + x)/(2a(x))\}$. Following the paper, the latency time is truncated at the 0.9 quantile of this Weibull distribution. Cured subjects are assigned infinite event times. Independent censoring times are then generated from an exponential distribution with mean 1.5. The observed time is defined as $Y = \min(T^*, C)$ with event indicator $\delta = I(T^* \leq C)$. We consider sample sizes $n = 50, 100, 200$.

Table 3.1 summarizes the rejection proportions of the incidence goodness-of-fit test statistic $\tilde{T}_n$ for different values of θ_2 and sample sizes n . The rejection proportion is defined as the proportion of the 500 simulated datasets for which the bootstrap p -value is less than $\alpha = 0.05$. The simulation setting fixes $\theta_0 = 1$, $\theta_1 = 1$, and $c_h = 1, 1.5, 2$, while two values of c_T are considered, where c_T affects the shape and scale of the Weibull latency distribution through the covariate-dependent parameters, and therefore changes the event-time behavior across covariate values. Each entry is based on 500 simulated datasets, and for each dataset the bootstrap p -value is obtained from 500 bootstrap samples.

Under the null hypothesis $\theta_2 = 0$, the nominal significance level is 0.05. To evaluate whether the observed rejection proportions are reasonably close to this level, the Monte Carlo standard error is

$$\sqrt{\frac{p(1-p)}{500}},$$

with $p = 0.05$, which gives approximately 0.0097. Thus, an approximate 95% Monte Carlo interval around 0.05 is

$$0.05 \pm 1.96(0.0097) \approx (0.031, 0.069).$$

Therefore, we considered observed rejection proportions in the $[0.032, 0.064]$ interval to be within the expected type I error range about our nominal 0.05 level. Based on this, the results suggest that the test maintains reasonable type I error control under the null hypothesis. As θ_2 increases, the rejection proportions also increase, and this pattern becomes more clear as the sample size grows. For example, when $c_T = 0$ and $\theta_2 = 2$, the rejection proportion increases from 18.2% for $n = 50$ to 73.0% for $n = 200$. Similar patterns are observed for $c_T = 1$. Therefore, these results indicate that the proposed test has increasing power as the model misspecification becomes larger and as the sample size increases.

Table 3.1: Rejection proportion (%) of the incidence GOF test statistic $\tilde{T}_n$ for different values of c_h , c_T , θ_2 , and sample size n . The simulation setting is $\theta_0 = 1$ and $\theta_1 = 1$. The null hypothesis corresponds to $\theta_2 = 0$, and the significance level is $\alpha = 0.05$.

c_h	c_T	θ_2	$n = 50$	$n = 100$	$n = 200$
1	0	0	3.6	6.4	4.6
		1	8.8	9.8	17.4
		2	19.0	38.8	69.6
		4	63.0	92.8	100.0
	1	0	4.0	3.8	6.2
		1	10.8	16.4	26.4
		2	23.4	46.6	74.0
		4	66.8	90.6	98.0
1.5	0	0	4.2	6.4	5.8
		1	9.6	12.0	18.8
		2	18.2	39.8	73.0
		4	63.4	93.8	100.0
	1	0	3.4	3.2	6.8
		1	10.0	16.2	27.6
		2	21.6	49.2	74.0
		4	66.4	90.6	98.6
2	0	0	3.2	6.4	5.0
		1	9.2	11.8	18.2
		2	18.6	38.6	73.6
		4	64.4	94.2	100.0
	1	0	2.4	3.4	5.2
		1	9.4	16.6	26.4
		2	19.8	46.4	74.0
		4	67.0	90.6	98.6

Chapter 4: Goodness-of-Fit Test for the Latency Component

4.1 Introduction

In a mixture cure model, the latency component describes the survival distribution among susceptible (uncured) subjects. In Chapter 2, this is represented by the conditional survival function $S_1(t \mid \mathbf{X})$. In this thesis, we assume a parametric form for this survival function, such as a Weibull distribution. However, if this parametric form is misspecified, then the fitted survival curves and interpretation for the susceptible group may be misleading. Therefore, besides checking the incidence component, it is also important to assess whether the assumed parametric latency model is appropriate. In this chapter, we describe the goodness-of-fit test proposed by Geerdens, Janssen, and Van Keilegom (2020) for the latency component. The test compares a nonparametric estimate of the susceptible survival function with the fitted parametric survival function, and then uses a parametric bootstrap procedure to compute the p -value [7].

4.2 Hypotheses

Let T denote the event time and C be the censoring time. We observe

$$\tilde{T} = \min(T, C), \quad \delta = I(T \leq C).$$

Under the mixture cure model, the conditional survival function is

$$S(t \mid \mathbf{X}) = (1 - \phi(\mathbf{X})) + \phi(\mathbf{X})S_1(t \mid \mathbf{X}), \quad (4.1)$$

where $\phi(\mathbf{X})$ denotes the probability of being susceptible given $\mathbf{X}$, and $S_1(t \mid \mathbf{X})$ is the survival function among susceptible subjects. The goal is to test whether the susceptible survival function belongs to a specified parametric family:

$$H_0 : S_1(t \mid \mathbf{X}) = S_1(t \mid \mathbf{X}; \theta) \quad \text{versus} \quad H_1 : S_1(t \mid \mathbf{X}) \neq S_1(t \mid \mathbf{X}; \theta), \quad (4.2)$$

for some parameter θ . $S_1(t \mid \mathbf{X}; \theta)$ may correspond to a Weibull distribution, or other distributions such as lognormal, Gompertz, and uniform.

4.3 Estimating the Susceptible Fraction

The procedure requires an estimate of the susceptible probability $\phi(\mathbf{X})$. In practice, we estimate this using the Kaplan–Meier estimator of the overall survival function. Let $\hat{S}(t)$ denote the Kaplan–Meier estimator of $S(t \mid \mathbf{X})$. Let

$$t_{1n} = \max\{\tilde{T}_i : \delta_i = 1\}$$

denote the largest observed uncensored event time. The plateau of the Kaplan–Meier curve provides an estimate of the cure probability, and thus the susceptible probability is estimated as

$$\hat{\phi} = 1 - \hat{S}(t_{1n}). \quad (4.3)$$

4.4 Nonparametric Estimation of the Susceptible Survival Function

From Equation (4.1), the susceptible survival function can be written as

$$S_1(t \mid \mathbf{X}) = \frac{S(t \mid \mathbf{X}) - (1 - \phi(\mathbf{X}))}{\phi(\mathbf{X})}.$$

Replacing $S(t | \mathbf{X})$ and $\phi(\mathbf{X})$ with their estimators gives

$$\hat{S}_{1,\text{nonpar}}(t) = \frac{\hat{S}(t) - (1 - \hat{\phi})}{\hat{\phi}}. \quad (4.4)$$

4.5 Fitting the Parametric Latency Model

Under the null hypothesis, the susceptible survival function follows $S_1(t | \mathbf{X}; \theta)$.

Using $\hat{\phi}$, the likelihood contribution for subject i is

If $\delta_i = 1$:

$$L_i = \hat{\phi} f_{1,\theta}(\tilde{T}_i),$$

If $\delta_i = 0$:

$$L_i = (1 - \hat{\phi}) + \hat{\phi} S_1(\tilde{T}_i | \mathbf{X}; \theta).$$

Combining these for n subjects, the log-likelihood is

$$\log L(\theta, \hat{\phi}) = \sum_{i=1}^n \left[\delta_i \log\{\hat{\phi} f_{1,\theta}(\tilde{T}_i)\} + (1 - \delta_i) \log\{1 - \hat{\phi} + \hat{\phi} S_1(\tilde{T}_i | \mathbf{X}; \theta)\} \right]. \quad (4.5)$$

The estimator is

$$\hat{\theta} = \arg \max_{\theta} \log L(\theta, \hat{\phi}).$$

The fitted survival function is

$$\hat{S}_{1,\text{param}}(t) = S_1(t | \mathbf{X}; \hat{\theta}).$$

4.6 Test Statistic

The latency goodness-of-fit test compares

$$\hat{S}_{1,\text{nonpar}}(t) \quad \text{and} \quad \hat{S}_{1,\text{param}}(t).$$

$$\Lambda_n = \sum_{i=1}^n \left\{ \hat{S}_{1,\text{nonpar}}(\tilde{T}_i) - \hat{S}_{1,\text{param}}(\tilde{T}_i) \right\}^2. \quad (4.6)$$

4.7 Bootstrap Procedure for the P -value

Similar to testing the goodness-of-fit using the incidence bootstrap procedure, we use a parametric bootstrap procedure under H_0 to assess the significance of the goodness-of-fit test statistic examining the latency component of the mixture cure model. The bootstrap procedure is summarized as follows:

1. Estimate $\hat{\phi}$ using Equation (4.3) and compute the observed test statistic Λ_{obs} using Equation (4.6). Then fit the parametric latency model under H_0 and obtain the estimated parameter vector $\hat{\theta}$.
2. Estimate the censoring survival function from the original data using the Kaplan–Meier estimator applied to $(\tilde{T}, 1 - \delta)$.
3. For $b = 1, \dots, B$, generate a bootstrap sample as follows:

(a) Generate

$$Z_{i,b}^* \sim \text{Bernoulli}(\hat{\phi}),$$

where $Z_{i,b}^* = 1$ indicates a susceptible subject and $Z_{i,b}^* = 0$ indicates a cured subject.

- (b) For subjects with $Z_{i,b}^* = 1$, generate bootstrap event times $T_{i,b}^*$ directly from the fitted parametric latency distribution under H_0 using the estimated parameter vector $\hat{\theta}$. If $Z_{i,b}^* = 0$, set $T_{i,b}^* = \infty$.
- (c) Generate bootstrap censoring times $C_{i,b}^*$ from the Kaplan–Meier estimate of the censoring distribution using inverse transform sampling on the Kaplan–Meier time grid. For each subject, generate $U_{i,b}^{(c)} \sim \text{Uniform}(0, 1)$ and let

$C_{i,b}^*$ be the first Kaplan–Meier time point such that

$$\hat{G}(t) \leq 1 - U_{i,b}^{(c)}.$$

If no such point exists, the largest Kaplan–Meier time point is used.

(d) Construct the bootstrap observed data

$$\tilde{T}_{i,b}^* = \min(T_{i,b}^*, C_{i,b}^*), \quad \delta_{i,b}^* = I(T_{i,b}^* \leq C_{i,b}^*).$$

(e) Recompute the latency goodness-of-fit test statistic from the bootstrap sample and denote it by Λ_b^* .

4. The bootstrap p -value is computed as

$$p\text{-value} = \frac{1}{B} \sum_{b=1}^B I(\Lambda_b^* \geq \Lambda_{\text{obs}}). \quad (4.7)$$

4.8 Simulation Results

In this simulation study, the susceptible indicator Y is generated from a Bernoulli distribution with probability ϕ , where ϕ denotes the susceptible fraction. For susceptible subjects, event times are generated from one of four candidate distributions: Weibull, Gompertz, lognormal, or uniform. Under the Weibull setting, the survival function is parameterized as $S_1(t) = \exp(-(\lambda t)^\rho)$. Cured subjects are assigned infinite event times. Independent censoring times are generated from a Weibull distribution with parameters λ_c and ρ_c . Specifically, we set $\lambda_c = 0.05$, and use $\rho_c = 1.5$ for the long plateau setting and $\rho_c = 3$ for the short plateau setting. These choices determine the amount of censoring and lead to different tail behaviors in the Kaplan–Meier curve. The observed time is defined as $\tilde{T} = \min(T, C)$ with event indicator $\delta = I(T \leq C)$.

We consider two scenarios: a long plateau setting with $\phi = 0.75$, representing a high susceptible rate, and a short plateau setting with $\phi = 0.25$, representing a low susceptible rate. Sample sizes $n = 150$ and $n = 500$ are used in both settings.

The results in Tables 4.1 and 4.2 summarize the performance of the latency goodness-of-fit test under the long and short plateau settings. Each result is based on 1000 simulated datasets, and for each dataset the bootstrap p -value is computed using 500 bootstrap samples. In both settings, the test maintains reasonable type I error control when the model is correctly specified. For example, when the true and fitted distributions are the same, the rejection rates at the 5% level are generally close to 0.05 for both sample sizes. When an incorrect distribution is fitted, the rejection rates increase substantially, especially as the sample size increases from $n = 150$ to $n = 500$. Therefore, these results suggest that the latency goodness-of-fit test performs reasonably well and has increasing power to detect model misspecification under both plateau settings.

The results in Tables 4.1 and 4.2 summarize the performance of the latency goodness-of-fit test under the long and short plateau settings. Each result is based on 1000 simulated datasets, and for each dataset the bootstrap p -value is computed using 500 bootstrap samples. Under the null hypothesis, when the true and fitted distributions are the same, the target rejection probability is the nominal significance level. To evaluate whether the observed rejection rates are reasonably close to the nominal level, the Monte Carlo standard error can be estimated by

$$\sqrt{\frac{p(1-p)}{1000}}.$$

At the 5% significance level, this gives

$$\sqrt{\frac{0.05(0.95)}{1000}} \approx 0.0069,$$

so an approximate 95% Monte Carlo interval around 0.05 is

$$0.05 \pm 1.96(0.0069) \approx (0.036, 0.064).$$

At the 10% significance level, the Monte Carlo standard error is

$$\sqrt{\frac{0.10(0.90)}{1000}} \approx 0.0095,$$

which gives an approximate 95% Monte Carlo interval around 0.10 of

$$0.10 \pm 1.96(0.0095) \approx (0.081, 0.119).$$

Therefore, rejection rates within these ranges can be viewed as consistent with simulation variability when the true type I error is equal to the nominal level. Based on this, the results suggest that the test maintains reasonable type I error control in both the long and short plateau settings when the model is correctly specified. When an incorrect distribution is fitted, the rejection rates increase substantially, especially as the sample size increases from $n = 150$ to $n = 500$. Therefore, these results suggest that the latency goodness-of-fit test performs reasonably well and has increasing power to detect model misspecification under both plateau settings.

Table 4.1: Simulation results for the latency goodness-of-fit test under a long plateau scenario ($\phi = 0.75$). Each result is based on 1000 simulated datasets with 500 bootstrap samples. Rejection rates (%) at the 5% and 10% significance levels are reported.

True Dist.	Fitted Dist.	n	Rej. Rate (5%)	Rej. Rate (10%)
Weibull	Weibull	150	4.7	10.0
	Gompertz	150	43.1	52.4
	Lognormal	150	81.7	87.2
	Uniform	150	99.6	99.7
	Weibull	500	5.2	8.8
	Gompertz	500	94.4	96.8
	Lognormal	500	99.9	100.0
	Uniform	500	100.0	100.0
Gompertz	Gompertz	150	4.8	9.8
	Weibull	150	54.9	66.0
	Lognormal	150	99.9	99.9
	Uniform	150	82.2	88.1
	Gompertz	500	5.4	10.7
	Weibull	500	98.3	99.0
	Lognormal	500	100.0	100.0
	Uniform	500	99.6	99.7
Lognormal	Lognormal	150	5.6	11.3
	Weibull	150	85.2	91.1
	Gompertz	150	100.0	100.0
	Uniform	150	99.8	100.0
	Lognormal	500	5.7	10.4
	Weibull	500	100.0	100.0
	Gompertz	500	100.0	100.0
	Uniform	500	100.0	100.0
Uniform	Uniform	150	4.9	9.8
	Weibull	150	96.1	98.5
	Gompertz	150	62.9	76.3
	Lognormal	150	100.0	100.0
	Uniform	500	5.4	9.6
	Weibull	500	100.0	100.0
	Gompertz	500	99.8	100.0
	Lognormal	500	100.0	100.0

Table 4.2: Simulation results for the latency goodness-of-fit test under a short plateau scenario ($\phi = 0.25$). Each result is based on 1000 simulated datasets with 500 bootstrap samples. Rejection rates (%) at the 5% and 10% significance levels are reported.

True Dist.	Fitted Dist.	n	Rej. Rate (5%)	Rej. Rate (10%)
Weibull	Weibull	150	5.3	9.2
	Gompertz	150	11.8	19.9
	Lognormal	150	19.9	33.1
	Uniform	150	82.3	87.7
	Weibull	500	4.5	10.2
	Gompertz	500	49.1	61.6
	Lognormal	500	85.9	90.9
	Uniform	500	100.0	100.0
Gompertz	Gompertz	150	4.3	9.7
	Weibull	150	15.2	25.3
	Lognormal	150	69.7	83.7
	Uniform	150	31.2	43.3
	Gompertz	500	5.8	10.4
	Weibull	500	62.6	73.8
	Lognormal	500	100.0	100.0
	Uniform	500	87.3	92.9
Lognormal	Lognormal	150	4.8	10.1
	Weibull	150	44.0	56.8
	Gompertz	150	84.0	90.2
	Uniform	150	79.3	88.3
	Lognormal	500	5.5	9.7
	Weibull	500	90.8	94.9
	Gompertz	500	100.0	100.0
	Uniform	500	99.7	100.0
Uniform	Uniform	150	4.9	9.1
	Weibull	150	43.0	54.8
	Gompertz	150	20.8	31.7
	Lognormal	150	79.4	92.8
	Uniform	500	4.8	9.7
	Weibull	500	98.5	99.4
	Gompertz	500	68.1	80.9
	Lognormal	500	100.0	100.0

Table 4.3: Comparison of rejection rates between the results reported in Geerdens et al. and the results obtained in this thesis. Rejection rates (%) at the 5% and 10% significance levels are shown under the long plateau ($\phi = 0.75$) and short plateau ($\phi = 0.25$) settings.

Setting	True	Fit	n	Paper (5%)	Paper (10%)	Thesis (5%)	Thesis (10%)
Long							
$(\phi = 0.75)$	Weibull	Weibull	150	4.2	10.1	4.7	10.0
		Weibull	500	4.5	8.9	5.2	8.8
		Gompertz	150	37.7	49.6	43.1	52.4
		Gompertz	500	83.1	89.5	94.4	96.8
		Lognormal	150	70.7	80.0	81.7	87.2
		Lognormal	500	100.0	100.0	99.9	100.0
		Uniform	150	99.9	100.0	99.6	99.7
		Uniform	500	100.0	100.0	100.0	100.0
Short							
$(\phi = 0.25)$	Weibull	Weibull	150	3.7	7.9	5.3	9.2
		Weibull	500	3.3	8.1	4.5	10.2
		Gompertz	150	6.9	14.6	11.8	19.9
		Gompertz	500	18.1	28.7	49.1	61.6
		Lognormal	150	1.7	5.7	19.9	33.1
		Lognormal	500	5.4	18.8	85.9	90.9
		Uniform	150	14.9	24.9	82.3	87.7
		Uniform	500	79.1	86.6	100.0	100.0

Chapter 5: Application to the Pediatric AML MLL Subgroup

5.1 Introduction

In this chapter, we fit a mixture cure model and apply the two goodness-of-fit tests described in Chapters 3 and 4 to the Pediatric AML MLL subgroup data to evaluate whether both the incidence and latency components are appropriate for this dataset.

5.2 Data Description

The application dataset comes from the Therapeutically Applicable Research to Generate Effective Treatments (TARGET) initiative from the National Cancer Institute [4]. In this thesis, we focus on a subset of pediatric AML patients with Mixed-Lineage Leukemia (MLL) rearrangements. After data cleaning, this dataset contains 290 patients and 17 variables. Among them, 133 patients experienced the event, and the overall survival time ranges from near zero to approximately 9.94 years. Among censored patients, the median follow-up time is 4.52 years with an interquartile range of 2.24 years. Since the Kaplan–Meier curve in Figure 1 shows a visible plateau

after several years and the follow-up period is sufficiently long, it is reasonable to investigate the existence of a cure fraction in this subgroup [4].

5.3 Mixture Cure Model Fitting

We fit a mixture cure model in which the incidence component models the probability that a patient is susceptible to the event, and the latency component models the survival distribution among susceptible patients.

In the main analysis, the model is fitted using the `flexsurvcure` package in R [8]. A Weibull distribution is assumed for the latency component. The covariates included in the model are age at diagnosis, log-transformed white blood cell count (`logWBC`), presence of central nervous system disease (`CNS.disease`), and bone marrow blast percentage (`bm.blast`). Survival time is converted to years for model fitting, and the event indicator equals 1 if the event is observed and 0 if the observation is censored.

For comparison, we also fit a mixture cure model using the `smcure` package [5], where the latency component follows a Cox proportional hazards model. The same set of covariates is included in both the incidence and latency components. This semi-parametric model is used as a reference to check whether the conclusions from the parametric Weibull latency model are consistent with a model that does not assume a specific parametric form for the latency distribution.

In addition to the Weibull latency model used in the main analysis, other parametric distributions, including lognormal, Gompertz, and exponential distributions, are also considered as sensitivity analyses to assess the robustness of the latency model assumption.

5.4 Application Results

Figure 5.1 shows the Kaplan–Meier survival curve for the Pediatric AML MLL subgroup. The survival curve drops relatively quickly during the first few years and then levels off after approximately 5 years. This plateau suggests that a proportion of patients may not experience the event during long-term follow-up, which supports the presence of a cure fraction and motivates the use of a mixture cure model.

Figure 5.2 shows the Kaplan–Meier curve overlaid with the fitted survival curves from the Weibull, Gompertz, lognormal, and uniform mixture cure models. From the plot, the Weibull and Gompertz models appear to follow the observed survival pattern reasonably well, while the lognormal and uniform models show poorer agreement, especially in the tail of the curve. This visual result is also consistent with the formal goodness-of-fit test results.

Table 5.1 presents the incidence goodness-of-fit test results for the logistic incidence model under two fitted cure models. For the parametric Weibull mixture cure model, the test statistic is 0.789 with a p -value of 0.066. Since this p -value is greater than 0.05, we fail to reject the null hypothesis, which suggests that the logistic incidence model provides an adequate fit in this setting. For the semiparametric Cox proportional hazards mixture cure model, the test statistic is 0.713 with a p -value of 0.040. Since this p -value is less than 0.05, the null hypothesis is rejected at the 5% level, which suggests some evidence of lack of fit for the logistic incidence specification under the semiparametric model. Therefore, under the parametric Weibull mixture cure model, the logistic incidence component appears to provide a reasonable fit to the data.

Table 5.2 presents the latency goodness-of-fit test results for several parametric latency distributions. Among the four fitted models, the Weibull model has the smallest test statistic (0.155), the largest bootstrap p -value (0.214), and the smallest AIC (721.185). Since the p -value is greater than 0.05, there is no evidence to reject the Weibull latency model, suggesting that it provides an adequate fit for the latency component. The Gompertz model gives a borderline result, with a p -value of 0.054, which is slightly above 0.05, so it is also not rejected at the 5% level. However, compared with the Weibull model, it has a larger test statistic and a slightly larger AIC, which suggests a weaker fit. In contrast, the lognormal and uniform models have very small p -values, indicating strong evidence against these two latency distributions. Their larger test statistics and larger AIC values also support this conclusion.

Figure 5.1: Kaplan–Meier survival curve for the Pediatric AML MLL subgroup. The yellow shaded region represents the 95% confidence interval, and tick marks indicate censored observations. The curve shows a plateau after approximately 5 years, suggesting the presence of a cure fraction.

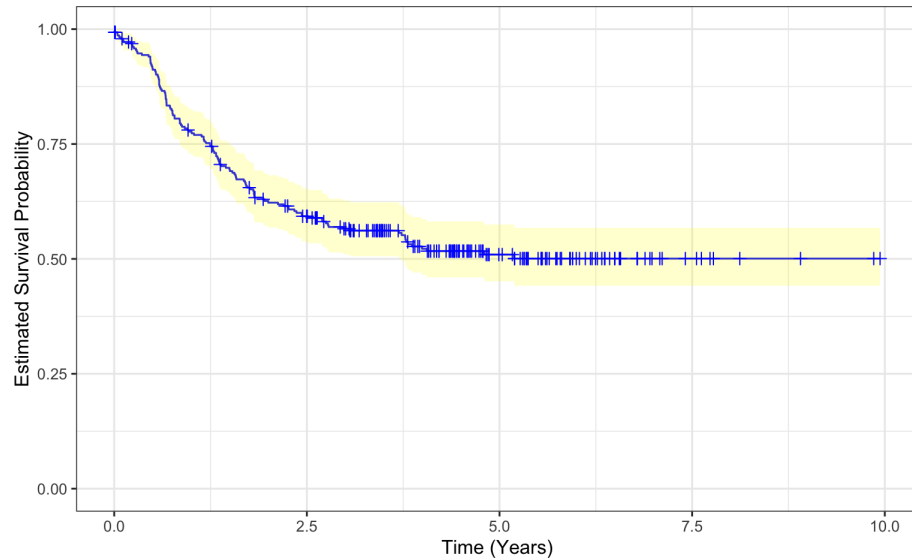


Figure 5.2: Kaplan–Meier survival curve for the Pediatric AML MLL subgroup overlaid with fitted survival curves under Weibull, Gompertz, lognormal, and uniform mixture cure models. Tick marks indicate censored observations.

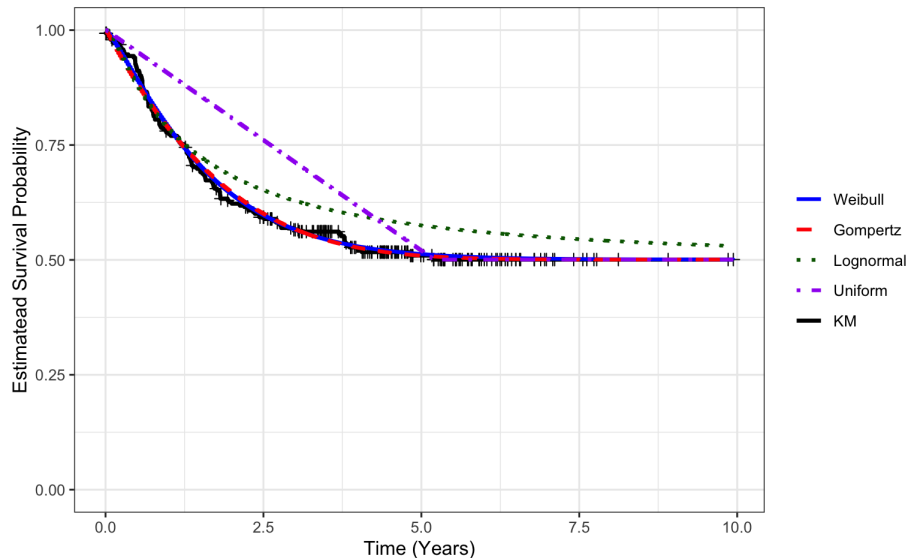


Table 5.1: Incidence goodness-of-fit test for the logistic incidence model under two mixture cure models applied to the Pediatric AML MLL data. Both models use a logistic specification for the incidence component.

Model	Test statistic	P-value
Parametric Weibull mixture cure model	0.789	0.066
Semiparametric Cox PH mixture cure model	0.713	0.040

Table 5.2: Results of the latency goodness-of-fit test applied to the Pediatric AML MLL data under different parametric latency distributions. The table reports the test statistic, bootstrap p -value, and the Akaike Information Criterion (AIC) for each fitted latency model.

	Weibull	Gompertz	Lognormal	Uniform
Test statistic	0.155	0.212	2.933	12.112
P-value	0.214	0.054	0.002	< 0.001
AIC	721.185	722.928	747.554	786.456

Chapter 6: Discussion

In this thesis, we studied goodness-of-fit tests for mixture cure models and examined how they can be used to assess both the incidence and latency components. In the simulation study for the incidence part, the test showed reasonable type I error control under the null hypothesis, and the rejection rates increased as the degree of misspecification became larger and as the sample size increased. For the latency part, the simulation results also showed reasonable performance under both the short and long plateau settings. When the fitted model was correctly specified, the rejection rates were generally close to the nominal levels. When the fitted model was misspecified, the rejection rates increased substantially, especially as the sample size increased from $n = 150$ to $n = 500$. Therefore, these results suggest that the latency goodness-of-fit test performs reasonably well and has increasing power to detect model misspecification under both plateau settings.

This thesis has several limitations. First, only four latency distributions were considered in the simulation study and real data application. Second, the real data application only used one dataset, so the results may not generalize well to other settings. In addition, although the updated latency simulation results are more reasonable, some differences from the results reported in Geerdens et al. still remain,

especially under certain misspecified settings, and these differences may be due to implementation details, bootstrap sampling, or parameterization choices.

There are also several directions for future study. One direction would be to further investigate why some latency simulation results still differ from those reported in Geerdens et al. Another direction would be to consider additional parametric or semiparametric latency models beyond the four distributions studied here. In addition, applying these goodness-of-fit tests to more real datasets would help assess whether the methods are stable and useful in practice. It would also be of interest to study an overall goodness-of-fit test for the full mixture cure model, rather than testing the incidence and latency components separately. Such a test could be useful for assessing the adequacy of the model as a whole. However, if the null hypothesis were rejected, it may be difficult to determine whether the misspecification comes from the incidence component, the latency component, or both, since a mixture cure model contains two linked components and many different model combinations are possible.

Appendix A: R Code for Incidence Goodness-of-Fit Test

A.1 Simulation Function

```
simulate_data <- function(n, theta = c(1, 1, 0), ct = 0,
  censor_mean = 1.5) {

  # Generate x from Uniform(-1, 1) as in Section 4.1
  x <- runif(n, -1, 1)
  x2 <- x^2

  # This is the uncured probability in the incidence model
  p_cured <- 1 / (1 + exp(theta[1] + theta[2] * x - theta
    [3] * x2))
  p_uncured <- 1 - p_cured

  # Z = 1 means uncured and Z = 0 means cured
  z <- rbinom(n, 1, p_uncured)

  # Weibull parameters for the latency part
  a_x <- (1 + x)^ct
  b_x <- exp(-(1 + x) / (2 * a_x))

  # Truncate the latency time at the 0.9 quantile like in
  # the paper
  tau <- qweibull(0.9, shape = a_x, scale = b_x)
  u <- runif(n)
  T_latency <- ifelse(
    u <= 0.9,
    qweibull(u, shape = a_x, scale = b_x),
    tau
  )

  # If someone is cured, their event time is infinite
```

```

T_star <- ifelse(z == 0, Inf, T_latency)

# Generate censoring times
C <- rexp(n, rate = 1 / censor_mean)

# Observed time and status
Y <- pmin(T_star, C)
delta <- as.integer(T_star <= C)

data.frame(Y = Y, delta = delta, x = x)
}

```

A.2 Nonparametric Estimation of Cure Probability

```

estimate_p_hat <- function(Y, delta, X, h) {
  # Use the same bandwidth for each subject
  h_est <- rep(h, length(Y))

  # Get the nonparametric estimate of the cure probability
  fit <- probcure(
    t = Y,
    d = delta,
    x = X,
    h = h_est,
    x0 = X
  )

  # Keep one value for each x
  fit$unique_q <- fit$q[!duplicated(fit$x0)]
  fit$unique_x <- fit$x0[!duplicated(fit$x0)]

  # Put p_hat back in the same order as X in the data
  p_hat <- fit$unique_q[match(X, fit$unique_x)]
  p_hat
}

```

A.3 Logistic Estimation

```

estimate_logistic <- function(x, y, eps = 1e-8) {
  # Add the intercept to the design matrix
  xmatrix <- as.matrix(cbind(Intercept = rep(1, length(y))
    , x))
}

```

```

theta <- rep(0, ncol(xmatrix))

# Pseudo-log-likelihood for the logistic model
neg_loglik <- function(theta) {
  pi_x <- 1 / (1 + exp(xmatrix %*% theta))

  # Keep pi_x away from 0 and 1
  pi_x <- pmin(pmax(pi_x, eps), 1 - eps)

  loglik <- sum(y * log(pi_x) + (1 - y) * log(1 - pi_x))
  -loglik
}

# Optimize the pseudo-log-likelihood
optim_result <- optim(
  par = theta,
  fn = neg_loglik,
  method = "BFGS",
  control = list(reltol = 1e-8)
)

# Return the estimated coefficients
optim_result$par
}

```

A.4 Test Statistic

```

compute_test_statistic <- function(p_hat, p_theta, h) {
  # Sample size
  n <- length(p_hat)

  # L2 distance between the nonparametric and parametric
  # cure probabilities
  l2_distance <- mean((p_hat - p_theta)^2)

  # Test statistic in the paper
  test_stat <- n * sqrt(h) * l2_distance

  test_stat
}

```

A.5 Incidence Goodness-of-Fit Test

```

incidence_gof <- function(time, status, x, B = 1000, ch =
  1.5) {

  n <- length(time)
  h0 <- ch * n^(-0.11)
  h <- ch * n^(-0.2)

  # Step 1: estimate the cure probability
  nonparametrically
  p_hat_vec <- estimate_p_hat(time, status, x, h = h0)

  # Step 2: fit the parametric logistic model
  theta_hat <- estimate_logistic(x, p_hat_vec)
  p_theta_hat <- 1 / (1 + exp(theta_hat[1] + theta_hat[2]
    * x))

  # Step 3a: estimate the event and censoring survival
  curves
  S_hat <- Tstar <- G_hat <- matrix(0, nrow = n, ncol = n)

  for (i in 1:n) {
    beran_event <- beran(
      x = x,
      t = time,
      d = status,
      x0 = x[i],
      h = h0,
      local = TRUE
    )

    # Save the common time grid from Beran
    if (i == 1) {
      test_time <- beran_event$testim
    }

    # S_hat is the conditional survival for the observed
    time
    S_hat[, i] <- unlist(beran_event$S)

    # G_hat is the conditional survival for the censoring
    time
    G_hat[, i] <- unlist(beran(
      x = x,
      t = time,
      d = 1 - status,

```



```

    x0 = x[i],
    h = h0,
    local = TRUE
  )$S)

# Turn S_hat into the susceptible survival curve Tstar
denom_i <- max(1 - p_hat_vec[i], 1e-8)
Tstar[, i] <- (S_hat[, i] - p_hat_vec[i]) / denom_i

# Keep the estimated curves inside [0, 1] and
# decreasing
Tstar[, i] <- pmin(pmax(Tstar[, i], 0), 1)
Tstar[, i] <- cummin(Tstar[, i])
G_hat[, i] <- pmin(pmax(G_hat[, i], 0), 1)
G_hat[, i] <- cummin(G_hat[, i])
}

test_statistic <- numeric(B)

for (b in 1:B) {
  # Generate susceptibility under the fitted null model
  Z <- runif(n) <= 1 - p_theta_hat

  # Generate bootstrap event times from the susceptible
  # survival curve
  T_b <- numeric(n)
  u <- runif(n)
  idx <- rep(NA_integer_, n)

  for (i in 1:n) {
    # Tstar is a survival curve, so use 1 - u in the
    # inverse step
    idx[i] <- which(Tstar[, i] <= 1 - u[i])[1]
  }

  T_b[is.na(idx)] <- max(test_time)
  T_b[!is.na(idx)] <- test_time[idx[!is.na(idx)]]
  T_b[!Z] <- Inf

  # Generate bootstrap censoring times from G_hat
  C_b <- numeric(n)
  u_c <- runif(n)
  idx_c <- rep(NA_integer_, n)

  for (i in 1:n) {

```

```

# Same inverse transform idea for the censoring
# survival curve
idx_c[i] <- which(G_hat[, i] <= 1 - u_c[i])[1]
}

C_b[is.na(idx_c)] <- max(test_time) + 1
C_b[!is.na(idx_c)] <- test_time[idx_c[!is.na(idx_c)]]

# Construct the bootstrap sample
Y_b <- pmin(T_b, C_b)
delta_b <- ifelse(T_b <= C_b, 1, 0)

# Recompute the test statistic on the bootstrap data
p_hat_vec_b <- estimate_p_hat(Y_b, delta_b, x, h = h)
theta_hat_b <- estimate_logistic(x, p_hat_vec_b)
p_theta_hat_b <- 1 / (1 + exp(theta_hat_b[1] + theta_hat_b[2] * x))
test_statistic[b] <- compute_test_statistic(p_hat_vec_b, p_theta_hat_b, h)
}

# Recompute the observed test statistic using h
p_hat_vec <- estimate_p_hat(time, status, x, h = h)
theta_hat <- estimate_logistic(x, p_hat_vec)
p_theta_hat <- 1 / (1 + exp(theta_hat[1] + theta_hat[2] * x))
observed_test <- compute_test_statistic(p_hat_vec, p_theta_hat, h)

# Bootstrap p-value
p_val <- mean(test_statistic >= observed_test)

list(
  test = observed_test,
  p_value = p_val,
  test_b = test_statistic
)
}

```

A.6 Simulation Study for the Incidence Goodness-of-Fit Test

```

# TABLE (INCIDENCE GOF)
# theta0 = 1, theta1 = 1

```

```

library(npcure)

theta0 <- 1
theta1 <- 1
theta2_vals <- c(0, 1, 2, 4)

n_vals <- c(50, 100, 200)

nsim <- 500
B_boot <- 500

results <- matrix(NA,
                  nrow = length(theta2_vals),
                  ncol = length(n_vals))

rownames(results) <- paste0("theta2=", theta2_vals)
colnames(results) <- paste0("n=", n_vals)

set.seed(42)

for (i in seq_along(theta2_vals)) {

  theta2_now <- theta2_vals[i]

  for (j in seq_along(n_vals)) {

    n_now <- n_vals[j]

    reject_vec <- numeric(nsim)

    for (sim in 1:nsim) {

      # ---- simulate data ----
      temp <- simulate_data(
        n = n_now,
        theta = c(theta0, theta1, theta2_now),
        ct = 1,                                     # plateau setting
        censor_mean = 1.5
      )

      time <- temp$Y
      status <- temp$delta
      x <- temp$x

      # incidence GOF

```

```

gof_out <- incidence_gof(
  time    = time,
  status  = status,
  x        = x,
  B        = B_boot,
  ch       = 1.5
)

# store rejection
reject_vec[sim] <- as.numeric(gof_out$p_value <
  0.05)
}

# average rejection rate
results[i, j] <- mean(reject_vec, na.rm = TRUE)

cat("done theta2 =", theta2_now, ", n =", n_now, "\n")
}
}

# round
results_round <- round(results, 3)

results_round

```

Appendix B: R Code for Latency Goodness-of-Fit Test

B.1 Simulation Function

```
simulate_parametric_data <- function(n, phi = 0.75,
                                     dist = "weibull",
                                     lambda = 0.50,
                                     lambda_c = 0.05,
                                     rho = 1.5,
                                     rho_c = 1.5,
                                     gompertz_shape = 0.75,
                                     gompertz_rate = 0.25,
                                     lognormal_meanlog = 0.50,
                                     lognormal_sdlog = sqrt
                                     (0.25),
                                     uniform_max = 3.25) {

  # Here phi is the susceptible fraction, so the cure fraction
  # is 1 - phi.
  Y <- rbinom(n, 1, phi)

  # Susceptible event times
  if (dist == "weibull") {
    # Using S(t)=exp(-(lambda*t)^rho)
    scale <- 1 / lambda
    T_i <- rweibull(n, shape = rho, scale = scale)
  } else if (dist == "gompertz") {
    T_i <- flexsurv::rgompertz(n, shape = gompertz_shape, rate
                              = gompertz_rate)
  } else if (dist == "lognormal") {
    T_i <- rlnorm(n, meanlog = lognormal_meanlog, sdlog =
                  lognormal_sdlog)
```

```

} else if (dist == "uniform") {
  T_i <- runif(n, min = 0, max = uniform_max)

} else {
  stop("Distribution not found")
}

# cured have infinite event time
T_i[Y == 0] <- Inf

# Censoring times
scale_c <- 1 / lambda_c
C_i <- rweibull(n, shape = rho_c, scale = scale_c)

T_tilde <- pmin(T_i, C_i)
delta <- as.integer(T_i <= C_i)

data.frame(T_tilde = T_tilde, delta = delta, Y = Y)
}

```

B.2 Estimate of the Susceptible Fraction

```

estimate_phi_hat <- function(T_tilde, delta) {

  fit <- survfit(Surv(T_tilde, delta) ~ 1)
  time <- fit$time
  S_hat <- fit$surv

  # Find the last survival probability before plateau
  if (sum(delta == 1, na.rm = TRUE) == 0) return(0)

  t1n <- max(T_tilde[delta == 1])
  j <- max(which(time <= t1n))
  if (!is.finite(j)) return(0)
  phi_hat <- 1 - S_hat[j]

  phi_hat
}

```

B.3 Nonparametric Latency Survival Estimate

```

estimate_S_hat_1 <- function(T_tilde, delta, phi_hat) {

  fit <- survfit(Surv(T_tilde, delta) ~ 1)
  S_hat <- fit$surv

  (phi_hat - 1 + S_hat) / phi_hat
}

```

B.4 Parametric Latency Model Fitting

```

fit_parametric_kja <- function(time, delta, phi_hat, dist = "
  weibull") {
  # basic clean (bootstrap can create NA sometimes)
  ok <- is.finite(time) & !is.na(time) & !is.na(delta)
  time <- pmax(time[ok], 1e-8)
  delta <- as.integer(delta[ok])

  # keep phi in (0,1)
  phi_hat <- min(max(phi_hat, 1e-6), 1 - 1e-6)

  eps <- 1e-12

  weibull <- function(theta, time = time, delta = delta, phi_
    hat = phi_hat) {
    lambda <- theta[1]
    rho <- theta[2]

    if (!is.finite(lambda) || !is.finite(rho) || lambda <= 0
      || rho <= 0) {
      return(1e12)
    }

    # Using:
    # f1(t)=lambda*rho*(lambda*time)^(rho-1)*exp(-(lambda*time
      )^rho)
    # S1(t)=exp(-(lambda*time)^rho)
    f1 <- lambda * rho * (lambda * time)^(rho - 1) * exp(-(
      lambda * time)^rho)
    S1 <- exp(-(lambda * time)^rho)

    f1 <- pmax(f1, eps)
    S1 <- pmin(pmax(S1, eps), 1 - eps)
  }
}

```

```

f1 <- f1[delta == 1]
S1 <- S1[delta == 0]

ll_1 <- sum(log(phi_hat) + log(f1))
ll_2 <- sum(log(1 - phi_hat + phi_hat * S1))

nll <- -(ll_1 + ll_2)
if (!is.finite(nll)) return(1e12)

nll
}

gompertz <- function(theta, time = time, delta = delta, phi_
  hat = phi_hat) {
  lambda <- theta[1]
  gamma <- theta[2]

  if (!is.finite(lambda) || !is.finite(gamma) || lambda <= 0
    || gamma <= 0) {
    return(1e12)
  }

  H <- lambda / gamma * (exp(gamma * time) - 1)
  if (any(!is.finite(H))) return(1e12)

  S1 <- exp(-H)
  f1 <- lambda * exp(gamma * time) * S1

  f1 <- pmax(f1, eps)
  S1 <- pmin(pmax(S1, eps), 1 - eps)

  f1 <- f1[delta == 1]
  S1 <- S1[delta == 0]

  ll_1 <- sum(log(phi_hat) + log(f1))
  ll_2 <- sum(log(1 - phi_hat + phi_hat * S1))

  nll <- -(ll_1 + ll_2)
  if (!is.finite(nll)) return(1e12)

  nll
}

uniform <- function(theta, time = time, delta = delta, phi_
  hat = phi_hat) {

```



```

theta <- theta[1]

if (!is.finite(theta) || theta <= 0) {
  return(1e12)
}

f1 <- ifelse(time < theta, 1 / theta, 0)
S1 <- ifelse(time < theta, 1 - time / theta, 0)

f1 <- f1[delta == 1]
S1 <- S1[delta == 0]

f1 <- pmax(f1, eps)
S1 <- pmin(pmax(S1, eps), 1 - eps)

ll_1 <- sum(log(phi_hat) + log(f1))
ll_2 <- sum(log(1 - phi_hat + phi_hat * S1))

nll <- -(ll_1 + ll_2)
if (!is.finite(nll)) return(1e12)

nll
}

lognormal <- function(theta, time = time, delta = delta, phi_hat = phi_hat) {
  mu <- theta[1]
  gamma <- theta[2] # variance

  if (!is.finite(mu) || !is.finite(gamma) || gamma <= 0) {
    return(1e12)
  }

  f1 <- 1 / (time * sqrt(2 * pi * gamma)) *
    exp(-((log(time) - mu)^2) / (2 * gamma))
  S1 <- 1 - pnorm((log(time) - mu) / sqrt(gamma))

  f1 <- pmax(f1, eps)
  S1 <- pmin(pmax(S1, eps), 1 - eps)

  f1 <- f1[delta == 1]
  S1 <- S1[delta == 0]

  ll_1 <- sum(log(phi_hat) + log(f1))
  ll_2 <- sum(log(1 - phi_hat + phi_hat * S1))

```

```

nll <- -(ll_1 + ll_2)
if (!is.finite(nll)) return(1e12)

nll
}

if (dist == "weibull") {
  est <- optim(
    par = c(0.5, 1.5), fn = weibull, time = time, delta =
      delta,
    phi_hat = phi_hat, method = "L-BFGS-B",
    lower = c(1e-12, 1e-12), upper = c(100, 100)
  )

  lambda_hat <- est$par[1]
  rho_hat <- est$par[2]

  theta_hat <- c(
    lambda = unname(lambda_hat),
    rho = unname(rho_hat)
  )

  S1_hat <- function(t) {
    pweibull(t, shape = rho_hat, scale = 1 / lambda_hat,
      lower.tail = FALSE)
  }

} else if (dist == "uniform") {
  est <- optim(
    par = max(time), fn = uniform, time = time, delta =
      delta,
    phi_hat = phi_hat, method = "L-BFGS-B", lower = 1e-12
  )

  theta_hat <- c(theta = unname(est$par[1]))

  S1_hat <- function(t) {
    punif(t, min = 0, max = theta_hat["theta"], lower.tail =
      FALSE)
  }

} else if (dist == "gompertz") {
  est <- optim(

```

```

    par = c(0.25, 0.75), fn = gompertz, time = time, delta =
      delta,
    phi_hat = phi_hat, method = "L-BFGS-B",
    lower = c(1e-12, 1e-12), upper = c(10, 10)
  )

  lambda_hat <- est$par[1]
  gamma_hat <- est$par[2]

  theta_hat <- c(
    rate = unname(lambda_hat),
    shape = unname(gamma_hat)
  )

  S1_hat <- function(t) {
    flexsurv::pgompertz(t, shape = gamma_hat, rate = lambda_
      hat, lower.tail = FALSE)
  }

} else if (dist == "lognormal") {
  mu0 <- mean(log(time), na.rm = TRUE)
  gamma0 <- var(log(time), na.rm = TRUE)

  est <- optim(
    par = c(mu0, gamma0), fn = lognormal, time = time, delta
      = delta,
    phi_hat = phi_hat, method = "L-BFGS-B",
    lower = c(-Inf, eps), upper = c(Inf, Inf)
  )

  mu_hat <- est$par[1]
  gamma_hat <- est$par[2]

  theta_hat <- c(
    mu = unname(mu_hat),
    gamma = unname(gamma_hat)
  )

  S1_hat <- function(t) {
    plnorm(t, meanlog = mu_hat, sdlog = sqrt(gamma_hat),
      lower.tail = FALSE)
  }

} else {

```

```

    stop("dist must be one of: weibull, gompertz, uniform,
          lognormal")
  }

  return(list(S1_hat = S1_hat, theta_hat = theta_hat, est =
             est))
}

```

B.5 Latency Goodness-of-Fit Test Statistic

```

latency_gof_test <- function(T_tilde, delta,
                             dist = "weibull") {

  # From step 2
  phi_hat <- estimate_phi_hat(T_tilde, delta)

  # Fit the overall KM survival
  km_fit <- survfit(Surv(T_tilde, delta) ~ 1)
  S_hat_at_T <- summary(km_fit,
                        times = T_tilde,
                        extend = TRUE)$surv

  # From step 2, estimate nonparametric latency survival
  S1_nonpar <- (phi_hat - 1 + S_hat_at_T) / phi_hat

  # From step 3, parametric latency survival model
  param_fit <- fit_parametric_kja(T_tilde,
                                  delta,
                                  phi_hat,
                                  dist)

  S1_param <- param_fit$S1_hat(T_tilde)

  sum((S1_nonpar - S1_param)^2,
      na.rm = TRUE)
}

```

B.6 Bootstrap Latency Goodness-of-Fit Test

```

bootstrap_latency <- function(time, status,
                              dist = "weibull",
                              B = 300) {

```

```

n <- length(time)

# Step 1: observed statistic
phi_hat <- estimate_phi_hat(time, status)
lambda_obs <- latency_gof_test(time, status, dist)

# fit parametric latency model under H0
param_fit <- fit_parametric_kja(time, status, phi_hat, dist)
theta_hat <- param_fit$theta_hat

# estimate censoring distribution G_hat via KM
ord <- order(time)
time_sorted <- time[ord]
status_sorted <- status[ord]

km_fit <- survfit(Surv(time_sorted, 1 - status_sorted) ~ 1)
tG <- km_fit$time
G_hat <- km_fit$surv

# generate susceptible event times directly from fitted
# parametric model
rlatency <- function(n, dist, theta_hat) {
  if (dist == "weibull") {
    lambda <- theta_hat["lambda"]
    rho <- theta_hat["rho"]
    rweibull(n, shape = rho, scale = 1 / lambda)

  } else if (dist == "gompertz") {
    flexsurv::rgompertz(
      n,
      shape = theta_hat["shape"],
      rate = theta_hat["rate"]
    )

  } else if (dist == "lognormal") {
    rlnorm(
      n,
      meanlog = theta_hat["mu"],
      sdlog = sqrt(theta_hat["gamma"])
    )

  } else if (dist == "uniform") {
    runif(
      n,
      min = 0,

```

```

        max = theta_hat["theta"]
    )

} else {
    stop("Unknown dist")
}
}

# sample censoring times from KM for censoring distribution
rcen_km <- function(n, tG, G_hat) {
    u_c <- runif(n)

    idx_c <- sapply(u_c, function(ui) {
        j <- which(G_hat <= 1 - ui)[1]
        if (is.na(j)) length(tG) else j
    })

    tG[idx_c]
}

# store bootstrap values
Lambda_b <- numeric(B)

# Step 2: bootstrap loop
for (b in seq_len(B)) {
    # 2.1 Generate cure indicator  $Y_i^b \sim \text{Bernoulli}(\text{phi\_hat})$ 
    Z <- rbinom(n, 1, phi_hat)

    # 2.2 Generate event times  $T_i^b$  from fitted  $S1\_hat$  for
    # susceptibles
    T_b <- rep(Inf, n)
    n_susc <- sum(Z == 1)
    if (n_susc > 0) {
        T_b[Z == 1] <- rlatency(n_susc, dist, theta_hat)
    }

    # 2.3 Generate censoring times  $C_i^b$  from KM estimate of
    # censoring distribution
    C_b <- rcen_km(n, tG, G_hat)

    # 2.4 Construct observed bootstrap data
    Y_b <- pmin(T_b, C_b)
    delta_b <- as.integer(T_b <= C_b)

    # 2.5 Bootstrap statistic

```

```

    Lambda_b[b] <- latency_gof_test(Y_b, delta_b, dist)
  }

  # Step 3: approximate p-value
  p_value <- mean(Lambda_b >= Lambda_obs)

  list(
    Lambda_obs = Lambda_obs,
    Lambda_b = Lambda_b,
    p_value = p_value
  )
}

```

B.7 Simulation Study for the Latency Goodness-of-Fit Test

```

# TABLE 4.1
# Long plateau: phi = 0.75 susceptible fraction

library(survival)
library(flexsurv)

# c("weibull", "gompertz", "lognormal", "uniform")
# c("gompertz", "weibull", "lognormal", "uniform")
# c("lognormal", "weibull", "gompertz", "uniform")
# c("uniform", "weibull", "gompertz", "lognormal")

set.seed(42)

phi_val <- 0.75 # long plateau: phi = 0.75, short plateau:
phi = 0.25
true_dist <- "uniform"
fit_dists <- c("uniform", "weibull", "gompertz", "lognormal")
n_vals <- c(150, 500)

nsim <- 1000
B_boot <- 500

table_long <- data.frame()

for (n_now in n_vals) {
  # store p-values
  pval_store <- matrix(NA, nrow = nsim, ncol = length(fit_
    dists))

```

```

colnames(pval_store) <- fit_dists

for (i in 1:nsim) {

  # generate one dataset
  temp <- simulate_parametric_data(
    n = n_now,
    phi = phi_val,
    dist = true_dist,
    lambda = 0.5,
    lambda_c = 0.05,
    rho = 1.5,
    rho_c = 1.5
  )

  time <- temp$T_tilde
  status <- temp$delta

  for (k in 1:length(fit_dists)) {

    fit_now <- fit_dists[k]

    # bootstrap GOF test
    gof_out <- bootstrap_latency(
      time,
      status,
      dist = fit_now,
      B = B_boot
    )

    # store p-value
    pval_store[i, k] <- gof_out$p_value
  }

  print(paste("done simulation", i, "for n =", n_now))
}

# compute rejection rates
for (k in 1:length(fit_dists)) {

  fit_now <- fit_dists[k]

  temp_row <- data.frame(
    true = "uniform",
    fit = fit_now,

```



```

      n = n_now,
      rej_5 = mean(pval_store[, k] < 0.05, na.rm = TRUE),
      rej_10 = mean(pval_store[, k] < 0.10, na.rm = TRUE)
    )

    table_long <- rbind(table_long, temp_row)
  }
}

# round results
table_long_round <- table_long
table_long_round[, 4:5] <- round(table_long_round[, 4:5], 3)

table_long_round

```

Appendix C: R Code for the Application to the MLL Dataset

C.1 Load Packages and Data

```
library(SummarizedExperiment)
library(survival)
library(tidyverse)
library(npcure)
library(ggplot2)
library(ggsurvfit)
library(smcurve)
library(flexsurvcure)

load("MLL.Rdata")
```

C.2 Kaplan–Meier Plot

```
km <- survfit(Surv(Overall.Survival.Time.in.Days, surv.ind) ~
  1, data = colData(MLL))
plot(km, mark.time = TRUE)

ggplot(
  summary(km, data.frame = TRUE, censored = TRUE) |>
    mutate(time_year = time / 365.24),
  aes(x = time_year, y = surv)
) +
  geom_step(color = "blue") +
  stat_stepribbon(aes(ymin = lower, ymax = upper), alpha =
    0.2, fill = "yellow") +
  geom_point(
    data = summary(km, data.frame = TRUE, censored = TRUE) |>
      mutate(time_year = time / 365.24) |>
      dplyr::filter(n.censor > 0),
```

```

aes(x = time_year, y = surv),
shape = 3, size = 2, color = "blue"
) +
coord_cartesian(ylim = c(0, 1)) +
labs(
  x = "Time (Years)",
  y = "Estimated Survival Probability",
  title = "Figure 1: Kaplan-Meier Survival Curve: TARGET-AML
          MLL Subgroup",
  subtitle = "The yellow shaded region represents the 95%
             confidence interval for the survival estimate."
) +
theme_bw()

```

C.3 Gene Expression Data Cleaning

```

expr <- assay(MLL)

zero_genes <- sum(apply(expr, 1, function(x) all(x == 0)))
nonzero_genes <- apply(expr, 1, function(x) !all(x == 0))

MLL_filtered <- MLL[nonzero_genes, ]

```

C.4 Phenotypic Data Cleaning

```

clinical_vars <- c(
  "race", "gender", "ethnicity", "Age.at.Diagnosis.in.Days", "
    age_at_index",
  "primary_diagnosis", "classification_of_tumor", "morphology"
  ,
  "year_of_diagnosis", "diagnosis_is_primary_disease",
  "vital_status", "First.Event",
  "Event.Free.Survival.Time.in.Days",
  "Overall.Survival.Time.in.Days",
  "Year.of.Last.Follow.Up",
  "surv.ind",
  "WBC.at.Diagnosis", "Bone.marrow.leukemic.blast.percentage
    ....",
  "Peripheral.blasts....", "CNS.disease", "Chloroma",
  "FLT3.ITD.positive.", "NPM.mutation", "CEBPA.mutation",
  "Risk.group", "Cytogenetically_Normal",
  "MRD.at.end.of.course.1", "MRD.at.end.of.course.2",

```

```

      "CR.status.at.end.of.course.1", "CR.status.at.end.of.course
      .2"
    )

pho <- colData(MLL_filtered)[, clinical_vars]
pho <- as.data.frame(pho)

na_df <- colSums(is.na(pho)) |>
  enframe(name = "variable", value = "NAs") |>
  filter(NAs > 0)

na_df

get_mode <- function(x) {
  output <- na.omit(x)
  output[which.max(tabulate(match(output, output)))]
}

pho_clean <- pho |>
  mutate(across(all_of(na_df$variable),
    ~ if (is.numeric(.x)) {
      replace(.x, is.na(.x), median(.x, na.rm =
        TRUE))
    } else {
      replace(.x, is.na(.x), get_mode(.x))
    }
  ))

head(pho_clean)
all(is.na(pho_clean))
dim(pho_clean)
names(pho_clean)

```

C.5 Final Analysis Dataset

```

df <- pho_clean |>
  mutate(
    Risk.group = factor(Risk.group),
    FLT3.ITD.positive. = factor(FLT3.ITD.positive.),
    CNS.disease = factor(CNS.disease),
    Chloroma = factor(Chloroma),
    gender = factor(gender),
    race = factor(race),
    ethnicity = factor(ethnicity)
  )

```

```

) |>
select(age_at_index, WBC.at.Diagnosis, Risk.group, FLT3.ITD.
       positive., CNS.disease, Chloroma,
       Bone.marrow.leukemic.blast.percentage...., Peripheral
       .blasts....,
       surv.ind, Overall.Survival.Time.in.Days) |>
mutate(
  surv.ind = as.integer(surv.ind),
  time_os  = as.numeric(Overall.Survival.Time.in.Days),
  age      = as.numeric(age_at_index),
  logWBC   = log1p(as.numeric(WBC.at.Diagnosis)),
  bm_blast = as.numeric(Bone.marrow.leukemic.blast.
                        percentage....),
  pb_blast = as.numeric(Peripheral.blasts....)
) |>
mutate(
  Risk.group = case_when(
    Risk.group == "Low Risk"      ~ 0L,
    Risk.group == "Standard Risk" ~ 1L,
    Risk.group == "High Risk"     ~ 2L
  ),
  FLT3.ITD.positive. = case_when(
    FLT3.ITD.positive. == "NO"      ~ 0L,
    FLT3.ITD.positive. == "UNKNOWN" ~ 1L,
    FLT3.ITD.positive. == "YES"     ~ 2L
  ),
  CNS.disease = case_when(
    CNS.disease == "No"      ~ 0L,
    CNS.disease == "Unknown" ~ 1L,
    CNS.disease == "Yes"     ~ 2L
  )
)

```

C.6 Data Summary

```

t_years <- df$time_os / 365.24

summary(t_years)
max(t_years, na.rm = TRUE)

cens_years <- t_years[df$surv.ind == 0]
summary(cens_years)

median(cens_years, na.rm = TRUE)

```

```
IQR(cens_years, na.rm = TRUE)
```

C.7 Model 1: Weibull Mixture Cure Model

C.7.1 Model Fitting

```
weibull_mcm <- flexsurvcure(  
  Surv(time_os / 365.24, surv.ind) ~ age + logWBC + CNS.  
    disease + bm_blast,  
  data = df,  
  mixture = TRUE,  
  dist = "weibull"  
)
```

C.7.2 Incidence Goodness-of-Fit Test

```
cf <- weibull_mcm$coefficients  
  
theta_hat <- c(  
  cf["theta"],  
  cf["age"],  
  cf["logWBC"],  
  cf["CNS.disease"],  
  cf["bm_blast"]  
)  
  
Z <- model.matrix(~ age + logWBC + CNS.disease + bm_blast,  
  data = df)  
x <- as.numeric(Z %*% theta_hat)  
  
set.seed(42)  
inc_out <- incidence_gof(  
  time = df$time_os / 365.24,  
  status = df$surv.ind,  
  x = x,  
  B = 500,  
  ch = 1.5  
)  
  
inc_out
```

C.7.3 Latency Goodness-of-Fit Test

```
set.seed(42)
lat_out <- bootstrap_latency(
  df$time_os / 365.24,
  df$urv.ind,
  dist = "weibull",
  B = 500
)
lat_out
```

C.8 Model 2: Semiparametric Mixture Cure Model

C.8.1 Model Fitting

```
fit <- smcure(
  Surv(time_os, surv.ind) ~ age + logWBC + CNS.disease + bm_
    blast,
  cureform = ~ age + logWBC + CNS.disease + bm_blast,
  data = df,
  model = "ph",
  Var = FALSE
)
```

C.8.2 Incidence Goodness-of-Fit Test

```
Z <- model.matrix(~ age + logWBC + CNS.disease + bm_blast,
  data = df)

cf <- fit$b

x_inc <- as.numeric(Z %*% cf)

set.seed(42)
inc_out <- incidence_gof(
  time = df$time_os / 365.24,
  status = df$urv.ind,
  x = x_inc,
  B = 500,
  ch = 1.5
)
```

```
inc_out
```


Bibliography

- [1] M. Amico and I. Van Keilegom. Cure models in survival analysis. *Annual Review of Statistics and Its Application*, 5(1):311–342, 2018.
- [2] J. Berkson and R. P. Gage. Survival curve for cancer patients following treatment. *Journal of the American Statistical Association*, 47(259):501–515, 1952.
- [3] J. W. Boag. Maximum likelihood estimates of the proportion of patients cured by cancer therapy. *Journal of the Royal Statistical Society: Series B (Methodological)*, 11(1):15–44, 1949.
- [4] H. Bolouri, J. E. Farrar, T. Triche, et al. The molecular landscape of pediatric acute myeloid leukemia reveals recurrent structural alterations and age-specific mutational interactions. *Nature Medicine*, 24:103–112, 2018.
- [5] C. Cai, Y. Zou, Y. Peng, and J. Zhang. smcure: An r-package for estimating semiparametric mixture cure models. *Computer Methods and Programs in Biomedicine*, 108(3):1255–1260, 2012.
- [6] Vernon T. Farewell. A model for a binary variable with time-censored observations. *Biometrika*, 64(1):43–46, 1977.
- [7] C. Geerdens, P. Janssen, and I. Van Keilegom. Goodness-of-fit test for a parametric survival function with cure fraction. *TEST*, 29(3):768–792, 2020.
- [8] C. H. Jackson. flexsurv: A platform for parametric survival modeling in r. *Journal of Statistical Software*, 70(8):1–33, 2016.
- [9] U. U. Müller and I. Van Keilegom. Goodness-of-fit tests for the cure rate in a mixture cure model. *Biometrika*, 106(1):211–227, 2019.
- [10] J. Xu and Y. Peng. Nonparametric cure rate estimation with covariates. *Canadian Journal of Statistics*, 42(1):1–17, 2014.